

Carlos Giménez

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GitHub: <https://github.com/carlosgl91>

Google Scholar: <https://scholar.google.es/citations?user=B53XXjIAAAAJ&hl>

Portfolio: carlosgl91.github.io/cgportfolio/

Full CV: https://github.com/carlosgl91/cg_cv_docs

EDUCATION

Diploma in Python for Data Science

Asunción, Paraguay

Universidad Comunera – 2021

GIS and Remote Sensing Specialization

Asunción, Paraguay

Universidad del Pacífico – 2019-2020

Obtained title: GIS and Remote Sensing Specialist. Thesis title: *Automated Mapping of COVID-19 Incidence and Population Mobility Variation in Paraguay. A R-Shiny based application*. Available at: <https://epi-gis-py.shinyapps.io/covid19py/>

Environmental Engineering Degree

San Lorenzo, Paraguay

National University of Asunción, College of Agricultural Sciences.

Obtained title: *Environmental Engineer*. Thesis title: *Analysis of the relationship between total forest biomass and NDVI values in the forest formations of the Cerrado, Concepción, Paraguay, summa cum laude*, 2019. Available at: github.com/carlosgl91/cg_bachelor_tesis

RECENT WORK EXPERIENCE

Climate Investment Partners LLC

Virginia, USA

Carbon projects technical assistance

Jan '22 – Present

- Carried out forest inventory data analysis for development of Reducing emissions from deforestation and forest degradation (REDD+) projects and automatization of this process (using R); Carried out monitoring carbon accounting for REDD+ and Afforestation, Reforestation, and Revegetation (ARR) projects for reporting to third party verifiers under the Verified Carbon Standard (VCS) from Verra.
- Carried out project baseline assessment, project due diligence (carbon laws, carbon ownership, land ownership, baseline land-use related laws and framework review).
- Carried out forest stratification and carbon stocks estimations for properties around the globe, using remote sensing datasets and field data.
- Assisted in forest inventory design for carbon stocks estimation within the framework of REDD+ project development.

Project Manager

- Lead project verification events accompanied by local partners and validation and verification bodies; Maintain active communication with local partners and third-party verifiers (validation and verification bodies).
- Drafted project description documents (Baseline scenario, eligibility, carbon accounting, leakage assessment, monitoring plan, country legal framework, risk assessment); Ensure adherence to relevant methodologies (e.g., Verra's VM0007 and AR-ACM0003 from the Clean Development Mechanism).
- Assisted in project origination and design across different countries with a focus on Paraguay, Brazil and Colombia.
- Led multidisciplinary teams including GIS/remote sensing experts, field staff, and local partners, coordinate field activities, data collection and stakeholder engagements.
- Developed and managed the execution of project workplans and timelines for REDD+ projects.

Remote sensing and GIS specialist

- Designed and carried out land use and landcover classification and accuracy assessments across multiple countries, ecosystems and land uses, for multiple type of projects and jurisdictions including Acre State of Brazil, Narok county in Kenya, West Sulawesi and Central Kalimantan in Indonesia, the Paraguayan Chaco, southern Ghana, and several countries in Central America, among others. Implemented multiple algorithm types such as SVM, Random Forest and CART, using tools and platforms like R, Google Earth Engine (GEE), Python, QGIS, ArcMap and ArcGIS Pro. Within the necessary workflow for imagery processing (cloud and shadow masking, void filling, speckle removal, etc.), preprocessed satellite imagery according to best practices such as Landsat constellations (5, 7, 8 and 9), Sentinel 2A and B, Sentinel 1, PLANET and high-resolution imagery such as SPOT and Pléiades.
- Designed, developed and implemented geoApps using Python, R and JavaScript (Leaflet and GEE) for different purposes, including automated spatial due diligence in the framework of the voluntary carbon markets (VCM), automated baseline estimation, project emissions and GHG credits eligible for issuance under the Verra-VCS standard, interactive classification results display, automated delineation of draft project areas for different project types (ARR, REDD+), interactive display of project spatial databases with dashboards for partners and investors.
- Designed, implemented and maintained spatial databases for multiple projects across several countries, utilizing POSTGRESQL, POSTGIS, Geoserver, and Geonode; performed EDAs, assessment and cleaning of large spatial and non-spatial datasets, creation of tidy datasets; ensured the consistency and integrity of datasets.
- Forest stratification and carbon stocks estimations for properties around the globe, using remote sensing datasets.

National University of Asuncion

San Lorenzo, Paraguay

Facultad Politécnica de la Universidad Nacional de Asunción, San Lorenzo Paraguay

March '25 - Present

“Artificial intelligence applied to the prediction of the spatio-temporal distribution of medically important vectors with a ‘One Health’ approach project - PINV01-528”. Project funded by the National Council of Science and Technology of Paraguay – CONACYT.

Associate researcher

- Generation of data derived from satellite imagery and auxiliary datasets to implement habitat suitability models.
- Programming geostatistical applications and tools for automating the application and evaluation of habitat suitability models (GEE, Python and R).
- Generation of tidy datasets based on spatio-temporal variables derived from satellite imagery (Sentinel 2) throughout the entire territory of the Republic of Paraguay.
- Capacity building, workshops about spatial datasets and applications of satellite imagery.
- Scientific paper writing.

Moisés Bertoni Foundation

Asunción, Paraguay

Fundación Moisés Bertoni

August '24 - Present

“Soundscapes: Applications of Artificial Intelligence for the Conservation and Use of Ecosystems - PINV01-530”. Project funded by the National Council of Science and Technology of Paraguay – CONACYT.

Associate researcher

- Developed potential distribution models for different taxa units classified from occurrence data, data derived from satellite imagery, climate models, landscape metrics and acoustic data.
- Programmed geostatistical applications and tools for automating the application and evaluation of habitat suitability models (GEE, Python and R).
- Applied and generated landscape ecology metrics, including spatially explicit layer generations for the spatial correlation with acoustic data.
- Processed satellite imagery data for statistical analysis and modelling.
- Delimited high priority conservation areas derived from indicators based on potential distribution models.
- Scientific paper writing.

SilviPar SA

Remote sensing and landscape specialist (consultancy)

Itapúa, Paraguay

August '23 – March '24

- Critical habitat analysis and proposal of net biodiversity balance. SilviPar
- Led a multidisciplinary team
- Annual mapping of land cover and use in 1,000,000 hectares in southern Paraguay (Ñeembucú, Itapúa, and Caazapá) with an emphasis on savanna formations and wetlands, from 2013 to 2023.
- Fragmentation and connectivity analysis and biological corridors proposal
- Fire incidence analysis including (fire hotspots over a decade over the different land use and landcover)
- Carried out a multitemporal classification based on Landsat 8 data; performed principal components analysis (PCA), segmentation (Mean-shift algorithm), Random Forest classifications using intra-annual seasonal variables, classification consistency analysis, accuracy assessments.
- Writing a scientific paper on the work

GENERAL SKILLS & INTERESTS

- Fluency in English (C1) and Spanish, intermediate proficiency in German (B1), Portuguese and Guaraní.
- Landscape ecology, wildlife conservation and forest carbon dynamics.
- Wildlife observation and photography; citizen science.
- Music, reading, chess, hiking, programming, languages, cinema.

COMPUTER SCIENCE SKILLS

- Advanced skills in GEE (JavaScript, Python APIs and R package); developing apps in JavaScript, Python, and R.
- Advanced knowledge of R programming and statistical language for both spatial and non-spatial data. Excellent skills with tidy and Shiny packages for developing both spatial and non-spatial applications.
- Advanced knowledge of Python and tools (e.g Conda, JupyterNotebook, PyQGIS, Geopandas, Rasterio, Geemap).
- Knowledge of the SQL language and PostgreSQL database engine with its POSTGIS extension.
- Knowledge of Linux Server; Docker; Git and the Github platform.
- Advanced knowledge of the xlsforms language and platforms such as Kobotoolbox.
- Advanced knowledge of the Geoserver, and Geonode platforms (installation and administration).
- Advanced knowledge of ArcGIS 10.8, Google Earth Pro, RStudio, PyQGIS, QGIS, Microsoft Office kit.
- Knowledge of the ArcGIS Online; ArcGIS Pro; Visual Studio Code (for Python, JavaScript, HTML and Mermaid).

PUBLICATIONS & TEACHING MATERIALS

Giménez Larrosa C. 2022. [Introduction to Geostatistics using R and QGIS](#). Classes developed (including presentations) within the framework of the Geostatistics Module of the Diploma in Data Science taught by the Polytechnic Faculty of Asunción.

Giménez Larrosa C. 2023. [Introduction to PyQGIS](#). Course taught by the Faculty of Agricultural Sciences of the National University of Asuncion.

Giménez, C. [Description of processes and methods used for the reclassification of 'Croplands' to "Land for livestock use" and "Land for agricultural use" throughout the national territory of Paraguay, periods 2000-2005-2011-2013-2015-2016](#). 2018. WWF-Paraguay.

Pérez-Estigarribia, [Proyecto TopaDengue](#) - Reporte Técnico 2.2.b - Informe final: TopaDengue Data Science - v2.0.0.

PE Pérez-Estigarribia, Velázquez Celeste Myriam, Giménez Larrosa, Carlos Antonio, Ramírez Pinto, Fredy. [Anfibios de la Reserva Natural Tapyta](#). Dpto Caazapá – Paraguay.

Del Castillo, H.; Velázquez, M.; Giménez, C.; [Distribution of threatened birds in protected wild areas in Paraguay](#). [First Paraguayan Zoology Congress](#). Asunción, Paraguay. November 25-29, 2019. Oral presentation.

Vera-Alcaraz HS; Pastor Pérez Estigarribia; Gilberto Antonio Benitez Rodas; Velázquez Myriam; Ramírez F; Giménez Carlos ; Rojas BL; Emhart J; Resquín JJ; [Comunidades de peces \(Actinopteri\) y anfibios \(Amphibia\) asociadas a bosque nativo y plantaciones de eucalipto en Caazapá, Paraguay](#). In: Congreso Paraguayo de Zoología 2019.

HS Vera-Alcaraz, PE Pérez-Estigarribia, GA Benítez, JE Emhart Vuletich , C. Rolón-Mendoza, C. Giménez Larrosa, M.C. Velázquez. 2017. [Aquatic communities associated with native forests and Eucalyptus plantations](#). spp. II Researchers' Meeting: Building Scientific Knowledge in Paraguay. Poster.

Kowalewski M, Velázquez M, Ramírez Pinto F, Coronel Molas C, Salvioni Recalde OD, Vega Gómez MC, Natalini MB, Pontón F, Martínez B, Miret J, Portillo L & Giménez Larrosa C. 2017. [Monitoring of mammals as indicators of the health status of ecosystems](#). III Paraguayan Conference on Herpetology and III Paraguayan Conference on Mastozoology. Category: Books and Summaries. Poster.